

Gaming Laptop

WHAT'S IN THE KIT

- Lenovo Legion 5 15" — i7-12700H, 16 GB RAM, 1 TB SSD, RTX 3050, Windows 11



QUICK-START GUIDE

1. Check out alongside tools that need horsepower (EinScan 3D scanner, video editing, Unity).
2. Charge fully; gaming laptops drain fast unplugged.
3. Create a student-use account/profile if multiple students will use it.

FULL LESSONS IN THIS GUIDE

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PRO TIP: This is the muscle behind the EinScan 3D scanner — consider checking them out together.

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LESSON · GRADES 6-8

Studio Workstation

LEARNING OUTCOMES

- Students will manage a shared production workflow with limited resources
- Students will prepare and queue rendering/export jobs efficiently
- Students will explain why some computing tasks need specialized hardware

MATERIALS

- Gaming laptop (the render station)
- Student Chromebooks for editing
- Shared folder for handoff files
- Render queue sign-up sheet

INSTRUCTIONS

1. Set up the laptop as the class 'render station' with a visible queue sheet.
2. Students edit video/3D projects on Chromebooks until export time.
3. Teach the handoff: file naming, shared folder, queue etiquette.
4. Students benchmark: time the same export on a Chromebook vs. the station.
5. Post the benchmark results — the 'why specialized hardware exists' lesson writes itself.
6. Rotate a student 'render tech' role to run the queue each session.

ACTIVITY

Studio Workstation — run a real production pipeline where heavy exports queue on the muscle machine. The Chromebook-vs-laptop benchmark makes computing power concrete.

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LESSON · GRADES 9-12

Game Design Bench

LEARNING OUTCOMES

- Students will navigate a professional game engine interface
- Students will build a simple interactive scene or prototype
- Students will evaluate game development as a career pathway

MATERIALS

- Gaming laptop with a free engine installed (Godot recommended; Unity/Unreal optional)
- Beginner tutorial (written or video)
- Design journal for documenting progress

INSTRUCTIONS

1. Install the engine and verify a template project runs before class.
2. Students rotate through in pairs following a starter tutorial (move a character, add a collectible).
3. Journal requirement each turn: what you built, what confused you, what you'd try next.
4. Milestone goal across sessions: one playable interaction per pair.
5. Career connection: match engine roles (design, art, code, audio) to real job postings.
6. Showcase: 90-second demos of each pair's scene.

ACTIVITY

Game Design Bench — hands on a real engine that Chromebooks can't run. Pairs ship one playable interaction and map the engine's disciplines to actual industry roles.